

^{13}C NUCLEAR MAGNETIC RESONANCE SPECTROSCOPY

BY

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The number of signals in ^{13}C NMR spectrum gives information that how many different kinds of carbons are present in that compound. Just as the number of signals in an NMR spectrum tells you how many different kinds of hydrogen a compound has. **The principles behind NMR and NMR spectroscopy are essentially the same.**

There are, however, some differences that make ^{13}C NMR easier to interpret.

1- ^{13}C NMR requires Fourier transform techniques because the signals obtained from a single scan are too weak to be distinguished from background electronic noise. However, FT- ^{13}C NMR scans can be repeated rapidly, so a large number of scans can be recorded and added.

Without Fourier transform, it could take days to record the number of scans required for a ^{13}C NMR spectrum.

2- **The individual signals are weak** because the isotope of carbon that gives rise to ^{13}C NMR signals constitutes only **1.11% of carbon atoms**. (The most abundant isotope of carbon, has no nuclear spin and therefore cannot produce an NMR signal.) The low abundance means that the intensities of the signals in ^{13}C NMR compared with those in ^1H NMR are reduced by a factor of approximately 100. In addition, **the gyromagnetic ratio (γ) of ^{13}C is about one-fourth that of ^1H and the intensity of a signal is proportional to γ^3** , Therefore, **the overall intensity of a ^{13}C signal is about 6400 times less than the intensity of an ^1H signal.**

3- One advantage to ^{13}C NMR spectroscopy is that the chemical shifts range over about 220 ppm, compared with about 12 ppm for ^1H NMR. This means that signals are less likely to overlap.

The NMR chemical shifts of different kinds of carbons are shown in Table below.

4- **The reference compound used in ^{13}C NMR is TMS**, the reference compound also used in ^1H NMR. Notice that **ketone and aldehyde carbonyl groups can be easily distinguished from other carbonyl groups.**

5- A disadvantage of ^{13}C NMR spectroscopy is that, unless special techniques are used, the area under a ^{13}C NMR signal is not proportional to the number of atoms giving rise to the signal. Thus, the number of carbons giving rise to a NMR signal cannot routinely be determined by integration.

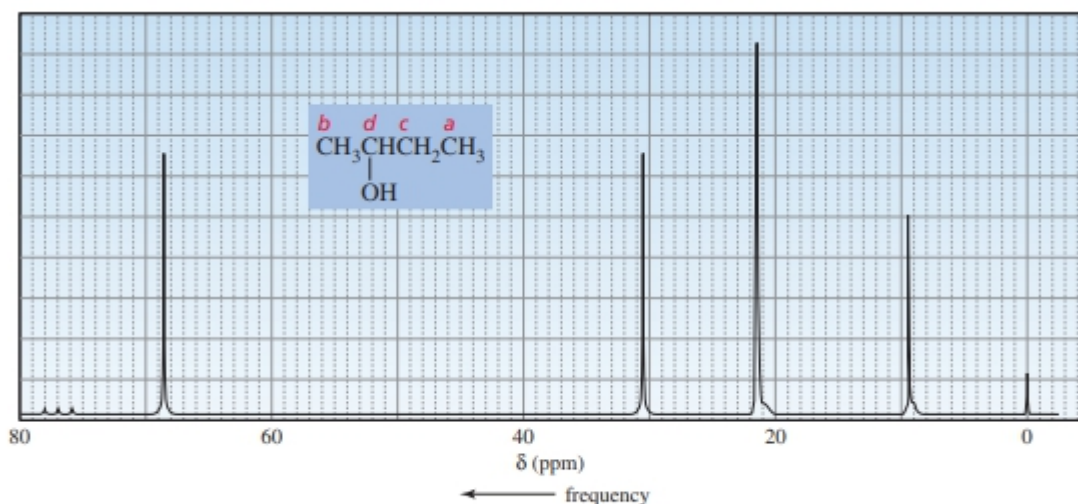
Type of carbon	Approximate chemical shift (ppm)	Type of carbon	Approximate chemical shift (ppm)
$(\text{CH}_3)_4\text{Si}$	0	$\text{C}-\text{I}$	0–40
$\text{R}-\text{CH}_3$	8–35	$\text{C}-\text{Br}$	25–65
$\text{R}-\text{CH}_2-\text{R}$	15–50	$\text{C}-\text{Cl}$	35–80
$\begin{array}{c} \text{R} \\ \\ \text{R}-\text{CH}-\text{R} \end{array}$	20–60	$\text{C}-\text{N}$	40–60
$\begin{array}{c} \text{R} \\ \\ \text{R}-\text{C}-\text{R} \\ \\ \text{R} \end{array}$	30–40	$\text{C}-\text{O}$	50–80
$\equiv\text{C}$	65–85	$\begin{array}{c} \text{R} \\ \diagup \\ \text{C}=\text{O} \\ \diagdown \\ -\text{N}- \end{array}$	165–175
$=\text{C}$	100–150	$\begin{array}{c} \text{R} \\ \diagup \\ \text{C}=\text{O} \\ \diagdown \\ \text{RO} \end{array}$	165–175
	110–170	$\begin{array}{c} \text{R} \\ \diagup \\ \text{C}=\text{O} \\ \diagdown \\ \text{HO} \end{array}$	175–185
		$\begin{array}{c} \text{R} \\ \diagup \\ \text{C}=\text{O} \\ \diagdown \\ \text{H} \end{array}$	190–200
		$\begin{array}{c} \text{R} \\ \diagup \\ \text{C}=\text{O} \\ \diagdown \\ \text{R} \end{array}$	205–220

The ^{13}C NMR spectrum of 2-butanol is shown in Figure 14.32. 2-Butanol has carbons in four different environments, so there are four signals in the spectrum.

The relative positions of the signals depend on the same factors that determine the relative positions of the proton signals in NMR. Carbons in electron-dense environments (Shielded) produce low-frequency signals (upfield), and carbons close to electron-withdrawing groups (deshielded) produce high-frequency signals (downfield).

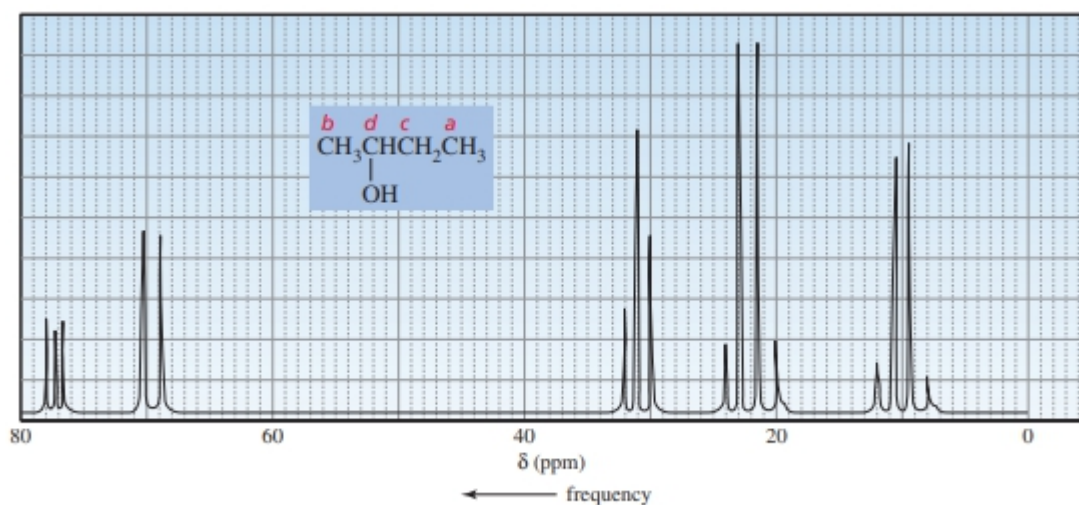
This means that the signals for the carbons of 2-butanol are in the same relative order that we would expect for the signals of the protons on those carbons in the ^1H NMR spectrum.

Thus, the carbon of the methyl group farther away from the electron-withdrawing OH group gives the lowest-frequency signal (upfield). As the frequency increases, the other methyl carbon comes next, followed by the methylene carbon; and the carbon attached to the OH group gives the highest-frequency signal (downfield).



▲ **Figure 14.32**
Proton-decoupled ^{13}C NMR spectrum of 2-butanol.

The signals are not normally split by neighboring carbons because there is little likelihood of an adjacent carbon being a ^{13}C . The probability of two carbons being next to each other is (about 1 in 10,000). (Because ^{12}C does not have a magnetic moment, it cannot split the signal of an adjacent ^{13}C .)



▲ **Figure 14.33**

The signals in a ^{13}C NMR spectrum can be split by nearby hydrogens. However, this splitting is not usually observed because the spectra are recorded using spin-decoupling, which obliterates the carbon-proton interactions. **Thus, all the signals are singlets in an ordinary NMR spectrum (Figure 14.32).**

If the spectrometer is run in a proton-coupled mode, the signals show spin-spin splitting. The splitting is not caused by adjacent carbons, but by the hydrogens bonded to the carbon that produces the signal.

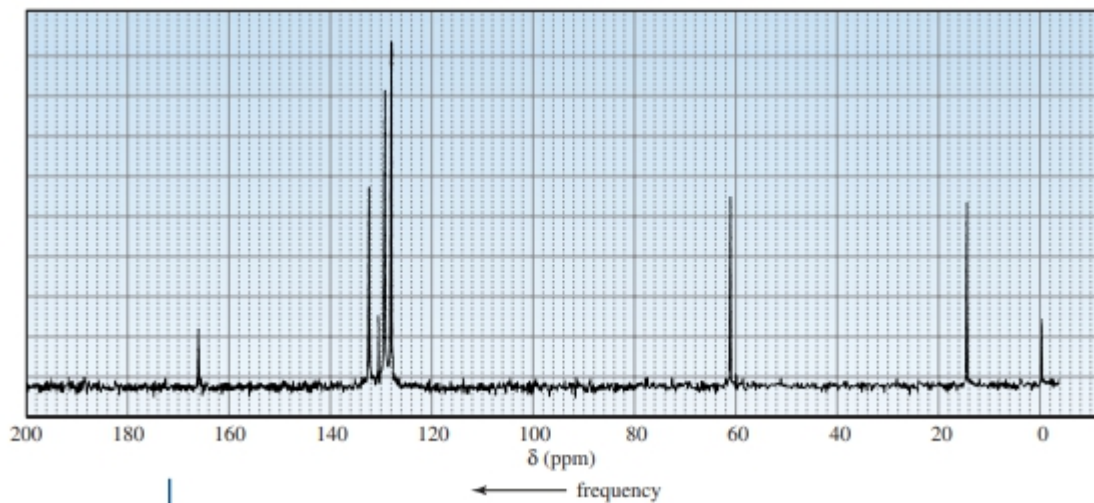
The multiplicity of the signal is determined by the $(n+1)$ rule.

The **proton-coupled ^{13}C NMR spectrum of 2-butanol** is shown in Figure 14.33. (The triplet at 78 ppm is produced by the solvent,) **The signals for the methyl carbons are each split into a quartet** because each methyl carbon is bonded to three hydrogens. **The signal for the methylene carbon is split into a triplet and the signal for the carbon bonded to the OH group is split into a doublet.**

The ^{13}C NMR spectrum of 2,2-dimethylbutane is shown in Figure 14.34.

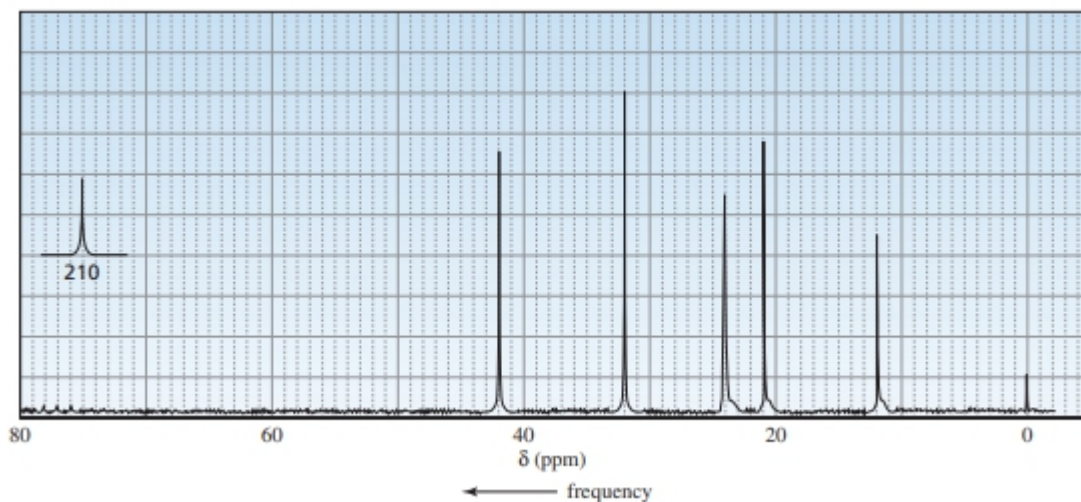
The three methyl groups at one end of the molecule are equivalent, so they all appear at the same chemical shift. The intensity of a signal is somewhat related to the number of carbons giving rise to the signal; consequently, the signal for these three methyl groups is the most intense signal in the spectrum. The tiny signal is for the Quaternary carbon; carbons that are not attached to hydrogens give very small signals. Because they are attached to a quaternary carbon, their signal appears at a higher frequency than that of the other methyl group (Table 14.4)

Problem3.1: Identify the compound with molecular formula $C_9H_{10}O_2$ that gives the following ^{13}C NMR spectrum:.

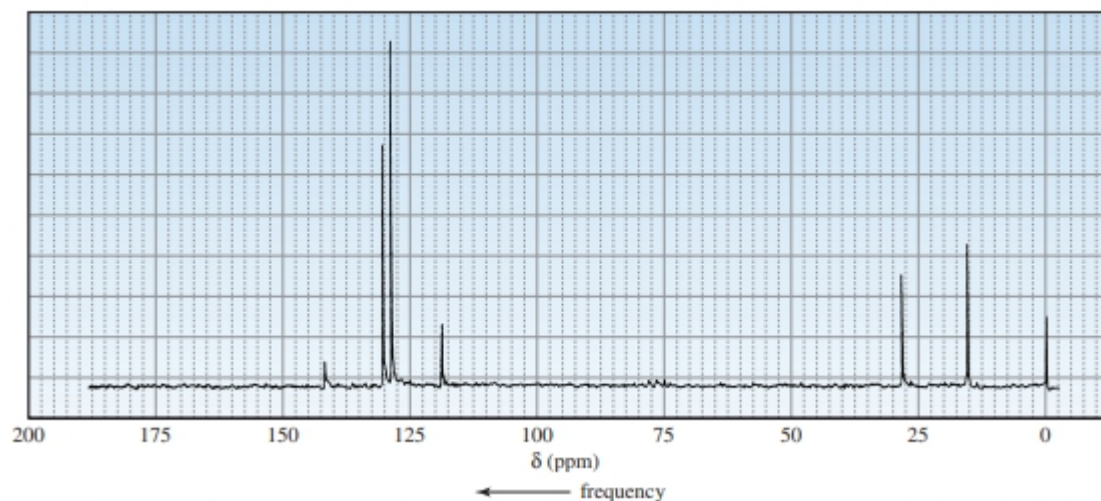


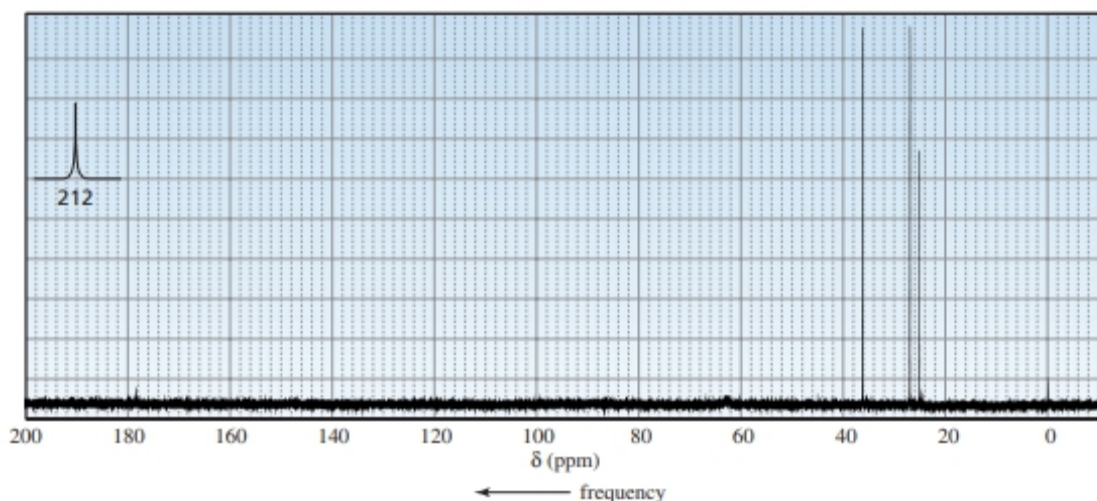
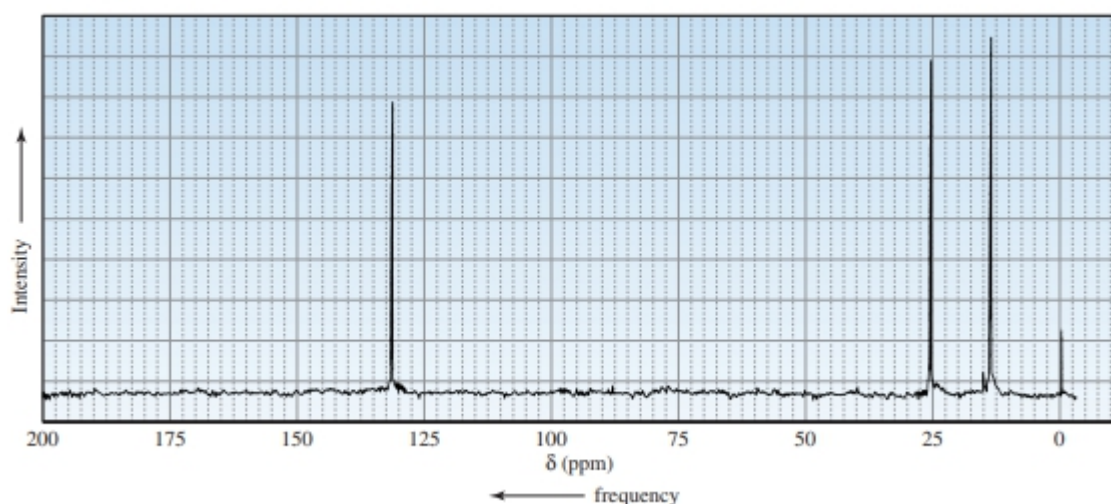
Problem3.2: Identify each compound in Figure 14.35 from its molecular formula and its ^{13}C NMR spectrum.

a. $C_{11}H_{22}O$



b. C_8H_9Br



c. $\text{C}_6\text{H}_{10}\text{O}$ d. C_6H_{12} 

DEPT NMR SPECTRA

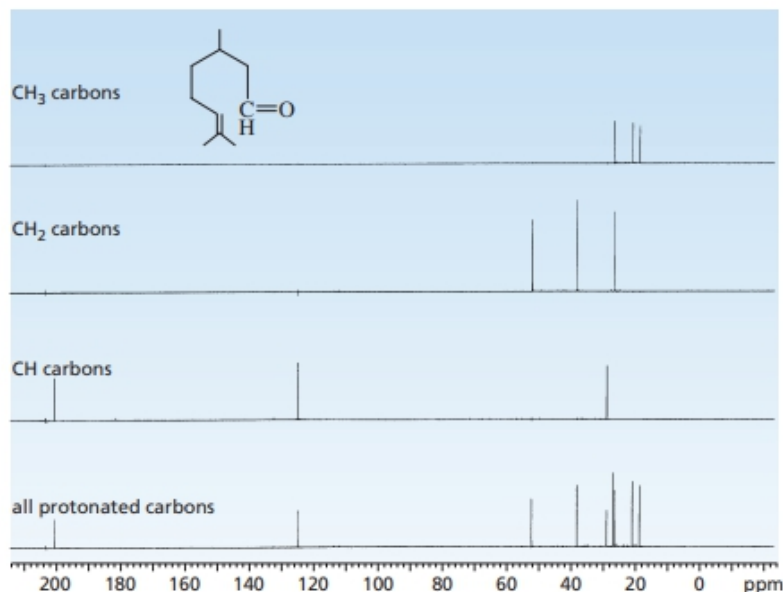
A technique called DEPT NMR has been developed to distinguish among C and CH groups. (DEPT stands for **distortionless enhancement by polarization transfer**.) It is now much more widely used than proton coupling to determine the number of hydrogens attached to a carbon.

The **DEPT ^{13}C NMR spectrum** of citronellal is shown in Figure 14.36. A DEPT ^{13}C NMR shows four spectra of the same compound.

The bottom most spectrum shows signals for all carbons that are covalently bonded to hydrogens. The next-to-the-bottom spectrum is run under conditions that allow only signals resulting from CH carbons to appear. The third spectrum is run under conditions that allow only signals produced by

CH₂ carbons to appear, and the top spectrum shows only signals produced by CH₃ carbons.

Figure 14.36 ▶
DEPT ¹³C NMR spectrum of
citronellal.



Information from the upper three spectra lets you determine whether a signal in a ¹³C NMR spectrum is produced by a CH₃, CH₂ or CH carbon.

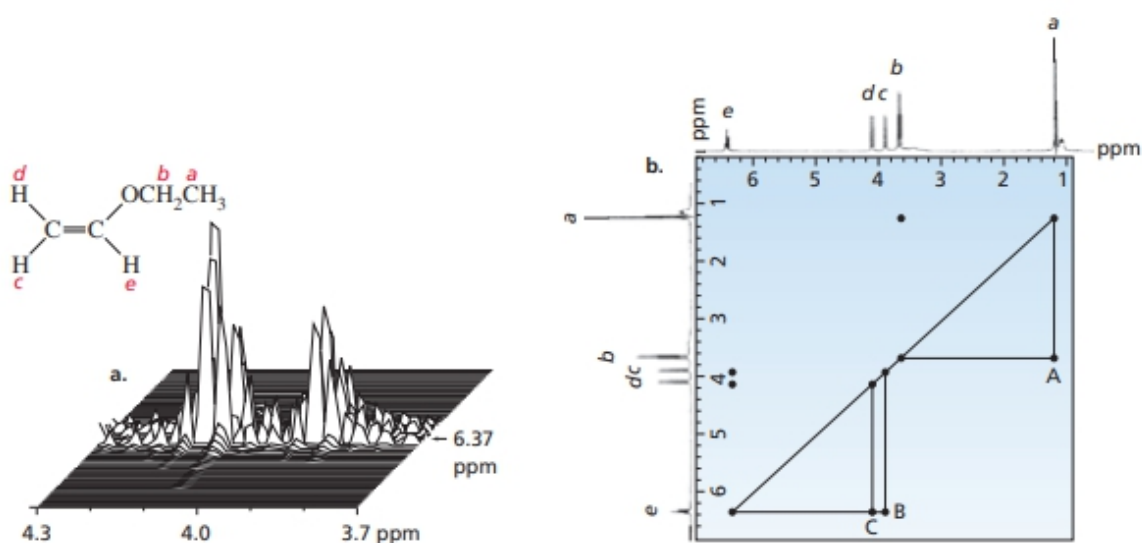
Notice that a DEPT NMR spectrum does not show a signal for a carbon that is not attached to a hydrogen. For example, the ¹³C NMR spectrum of 2-butanone shows four signals because it has four nonequivalent carbons, whereas the DEPT NMR spectrum of 2-butanone shows only three signals because the carbonyl carbon is not bonded to a hydrogen, so it will not produce a signal.

Two-Dimensional NMR Spectroscopy

Complex molecules such as proteins and nucleic acids are difficult to analyze by NMR because the signals in their NMR spectra overlap. Such compounds are now being analyzed by two-dimensional (2-D) NMR. Unlike X-ray crystallography, 2-D NMR techniques allow scientists to determine the structures of complex molecules in solution. Such a determination is particularly important for biological molecules whose properties depend on how they fold in water. More recently, 3-D and 4-D NMR spectroscopy have been developed and can be used to determine the structures of highly complex molecules. A thorough discussion of 2-D NMR is beyond the scope of this book, but the discussion that follows will give you a brief introduction to this increasingly important spectroscopic technique. The ¹H NMR and

^{13}C NMR spectra discussed in the preceding sections have one frequency axis and one intensity axis; 2-D NMR spectra have two frequency axes and one intensity axis. The most common 2-D spectra involve shift correlations; they identify protons that are coupled (i.e., that split each other's signal). This is called shift-correlated spectroscopy, which is known by the acronym COSY.

A portion of the COSY spectrum of ethyl vinyl ether is shown in Figure 14.37(a); it looks like a mountain range viewed from the air because intensity is the third axis. These “mountain-like” spectra (known as stack plots) are not the spectra actually used to identify a compound. Instead, the compound is identified using a contour plot (Figure 14.37(b)), where each mountain in Figure 14.37(a) is represented by a large dot (as if its top had been cut off). The two mountains shown in Figure 14.37(a) correspond to the dots labeled B and C in Figure 14.37(b).



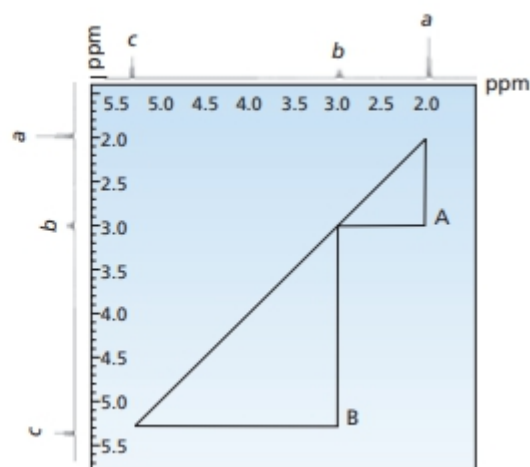
▲ **Figure 14.37**

(a) COSY spectrum of ethyl vinyl ether (stack plot). (b) COSY spectrum of ethyl vinyl ether (contour plot). In a COSY spectrum, an ^1H NMR spectrum is plotted on both the x- and y-axes. Cross peaks B and C represent the mountains in (a).

In the contour plot (Figure 14.37(b)), the usual one-dimensional NMR spectrum is plotted on both the x- and y-axes. To analyze the spectrum, a diagonal line is drawn through the dots that bisect the spectrum. The dots that are not on the diagonal (A, B, C) are called cross peaks. Cross peaks indicate pairs of protons that are coupled. For example, if we start at the cross peak labeled A and draw a straight line parallel to the y-axis back to

the diagonal, we hit the dot on the diagonal at produced by the protons. If we next go back to A and draw a straight line parallel to the x-axis back to the diagonal, we hit the dot on the diagonal at produced by the protons. This means that the and protons are coupled. If we then go to the cross peak labeled B and draw two perpendicular lines back to the diagonal, we see that the and protons are coupled; the cross peak labeled C shows that the and protons are coupled. Notice that we used only cross peaks below the diagonal; the cross peaks above the diagonal give the same information. Notice also that there is no cross peak due to the coupling of and consistent with the absence of coupling for two protons bonded to an carbon in the one-dimensional NMR spectrum shown in Figure 14.17. The COSY spectrum of 1-nitropropane is shown in Figure 14.38. Cross peak A shows that the and protons are coupled, and cross peak B shows that the and protons are coupled. Notice that the two triangles in the figure have a common vertex, since the protons are coupled to both the and protons.

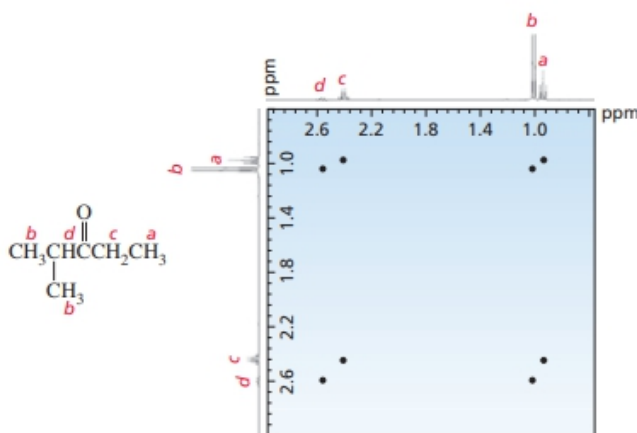
Figure 14.38 ►
COSY spectrum of 1-nitropropane.



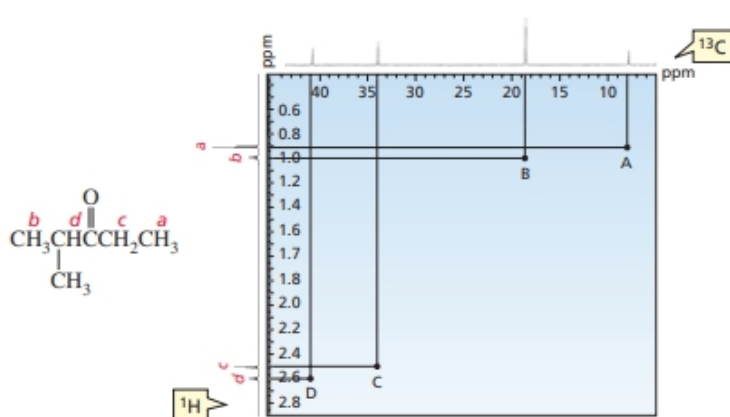
PROBLEM 40

Identify pairs of coupled protons in 2-methyl-3-pentanone, using the COSY spectrum in Figure 14.39.

Figure 14.39 ▶
COSY spectrum for Problem 40.



2-D NMR spectra that show shift correlations are called HETCOR (from heteronuclear correlation) spectra. HETCOR spectra indicate coupling between protons and the carbon to which they are attached. The HETCOR spectrum of 2-methyl-3-pentanone is shown in Figure 14.40. The NMR spectrum is shown on the x-axis and the NMR spectrum is shown on the y-axis. The cross peaks in a HETCOR spectrum identify which hydrogens are attached to which carbons. For example, cross peak A indicates that the hydrogens that show a signal at in the NMR spectrum are bonded to the carbon that shows a signal at in the NMR spectrum. Cross peak C shows that the hydrogens that show a signal at are bonded to the carbon that shows a signal at '34 ppm.



◀ **Figure 14.40**
HETCOR spectrum of 2-methyl-3-pentanone. A HETCOR spectrum indicates coupling between protons and the carbons to which they are attached.

Clearly, 2-D NMR techniques are not necessary for assigning the signals in the NMR spectrum of a simple compound such as 2-methyl-3-pentanone. However, in the case of many complicated molecules, signals can be

assigned only with the aid of 2-D NMR. Spectra showing shift correlations (called 2-D INADEQUATE spectra) identify directly bonded carbons. There also are spectra in which chemical shifts are plotted on one of the frequency axes and coupling constants on the other. Other 2-D spectra involve the nuclear Overhauser effect (NOESY for very large molecules, ROESY for mid-size molecules).³ These spectra are used to locate protons that are close together in space.

Magnetic Resonance Imaging NMR has become an important tool in medical diagnosis because it allows physicians to probe internal organs and structures without invasive surgical methods or the harmful ionizing radiation of X-rays. When NMR was first introduced into clinical practice in 1981, the selection of an appropriate name was a matter of some debate. Because some members of the public associate nuclear processes with harmful radiation, the “N” was dropped from the medical application of NMR, now known as magnetic resonance imaging (MRI). The spectrometer is called an MRI scanner. An MRI scanner consists of a magnet sufficiently large to accommodate an entire patient, along with additional coils for exciting the nuclei, modifying the magnetic field, and receiving signals. Different tissues yield different signals, which are separated into components by Fourier transform analysis. Each component can be attributed to a specific site of origin in the patient, allowing a cross-sectional image of the patient’s body to be constructed. ¹³C ¹³C–¹³C 3 NOESY and ROESY are the acronyms for nuclear Overhauser effect spectroscopy and rotationframe Overhauser effect spectroscopy.

An MRI may be taken in any plane, essentially independently of the patient’s position, allowing optimum visualization of the anatomical feature of interest. By contrast, the plane of a computed tomography (CT) scan, which uses X-rays, is defined by the position of the patient within the machine and is usually perpendicular to the long axis of the body. CT scans in other planes may be obtained only if the patient is a skilled contortionist. Most of the signals in an MRI scan originate from the hydrogens of water molecules because these hydrogens are far more abundant in tissues than are the hydrogens of organic compounds. The difference in the way water is bound

in different tissues is what produces much of the variation in signal among organs, as well as the variation between healthy and diseased tissue (Figure 14.41). MRI scans, therefore, can sometimes provide much more information than images obtained by other means. For example, MRI can provide detailed images of blood vessels. Flowing fluids such as blood respond differently to excitation in an MRI scanner than do stationary tissues and normally do not produce a signal. However, the data may be processed to eliminate signals from motionless structures, thereby showing signals only from the fluids. This technique is sometimes used instead of more invasive methods to examine the vascular tree. It is now possible to completely suppress the signal from certain types of tissue (usually fat). It is also possible to differentiate intracellular and extracellular edema, which is important in assessing patients suspected of having suffered strokes. The versatility of MRI has been enhanced by the use of gadolinium as a contrast reagent. Gadolinium, a paramagnetic metal, modifies the magnetic field in its immediate vicinity, altering the signal from hydrogen nuclei in close proximity. The distribution of gadolinium, which is infused into a patient's veins, may be modified by certain disease processes, such as cancer and inflammation. These abnormal distributions appear in the images obtained from appropriate scanning techniques. NMR spectroscopy using is not yet in routine clinical use, but is being used widely in clinical research. It is of particular interest because ATP and ADP (Sections 17.20 and 27.2) are involved in most metabolic processes, so it will provide a way to investigate cellular metabolism.